D 91713 (Pages : 4) Name.....

Reg. No.....

THIRD SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION NOVEMBER 2020

Mathematics

MAT 3B 03—CALCULUS AND ANALYTIC GEOMETRY

Time : Three Hours Maximum : 80 Marks

Part A

Answer all the **twelve** *questions.*
Each question carries 1 *mark.*

1. Evaluate $\int_0^{\pi/6} \tan 2x \, dx$.

2. Define an alternating series.

3. Find the vertices of the hyperbola $\frac{y^2}{4} - \frac{x^2}{5} = 1$.

4. Find the Taylor polynomial of order zero generated by $f(x) = \sin x$ at $a = \frac{\pi}{4}$.

5. Evaluate $\frac{d}{dt}\left(\tan h\sqrt{1+t^2}\right)$.

6. Examine whether $\sum_{n=1}^{\infty} n^2$ converges or diverges.

7. Find the directive of the parabola $y^2 = 10x$.

8. Define absolute convergence.

9. Find y if $\ln y = 3t + 5$.

10. Find the parametric equation of the circle $x^2 + y^2 = a^2$.

Turn over

11. Examine whether $x^2 + xy + y^2 - 1 = 0$ represents a parabola, ellipse or hyperbola.

12. Evaluate $\lim_{x \to 0} \frac{\sqrt{1+x} - 1}{x}$.

(12 × 1 = 12 marks)

Part B

Answer any **nine** *questions.*
Each question carries 2 marks.

13. For what values of x do the series $\sum_{n=0}^{\infty} n!\, x^n$ converges.

14. Evaluate $\frac{d}{dx} \ln_{10}(3x+1)$.

15. Evaluate $\int \frac{\log_2 x}{x}\, dx$.

16. Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $a = 2$.

17. Find $\frac{dy}{dx}$ if $y = x^x, x > 0$.

18. Graph the set of points whose polar co-ordinates satisfy the conditions $1 \le r \le 2$ and $0 \le 0 \le \frac{\pi}{2}$.

19. Examine whether the series :

$$5 + \frac{2}{3} + 1 + \frac{1}{7} + \frac{1}{2} + \frac{1}{3!} + \frac{1}{4!} + \ldots + \frac{1}{k!} + \ldots \text{ converges.}$$

20. Evaluate $\int_0^{\ln 2} 4e^x \sin h x\, dx$.

21. Show that $\ln x$ grows slaver than x as $x \to \infty$.

22. Examine whether $\sum_{n=1}^{\infty}(-1)^{n+1}$ converges or diverges.

23. Evaluate $\lim_{x\to 0}\frac{3x-\sin x}{x}$.

24. Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\frac{4\cos\theta}{3+2\sin\theta}d\theta$.

(9 × 2 = 18 marks)

Part C

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. The standard parametrization of the circle of radius 1 centered at the point (0, 1) in the xy-plane is $x=\cos t, y=1+\sin t, 0\le t\le 2\pi$. Use the parametrization to find the area of the surface swept out by revolving the circle about the x-axis.

26. Find the centroid of the first quadrant of the astroid $x=\cos^3 t, y=\sin^3 t, 0\le t\le 2\pi$.

27. Show that $\lim_{x\to 0^+}(1+x)^{\frac{1}{x}}=e$.

28. Using Integral test show that the p-series :

$$\sum_{n=1}^{\infty}\frac{1}{n^p}=\frac{1}{1^p}+\frac{1}{2^p}+\ldots+\frac{1}{n^p}+\ldots \text{ converges if } p>1 \text{ and diverges if } p\le 1.$$

29. Find the Taylor polynomial generated by $f(x)=\cos x$ at $x=0$.

30. Find the length of the Cardioid $r=1-\cos\theta$.

31. Prove that if $\sum_{n=1}^{\infty}|a_n|$ converges then $\sum_{n=1}^{\infty}a_n$ converges.

32. Graph the curve $r^2 = 4\cos\theta$.

33. Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,n!}$.

(6 × 5 = 30 marks)

Part D

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Find the area of the region in the plane enclosed by the Cardioid $r = 2(1+\cos\theta)$.

35. Show that the Maclaurin's series for $\cos x$ converges to $\cos x$ for every value of x.

36. Using Integral test examine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

(2 × 10 = 20 marks)